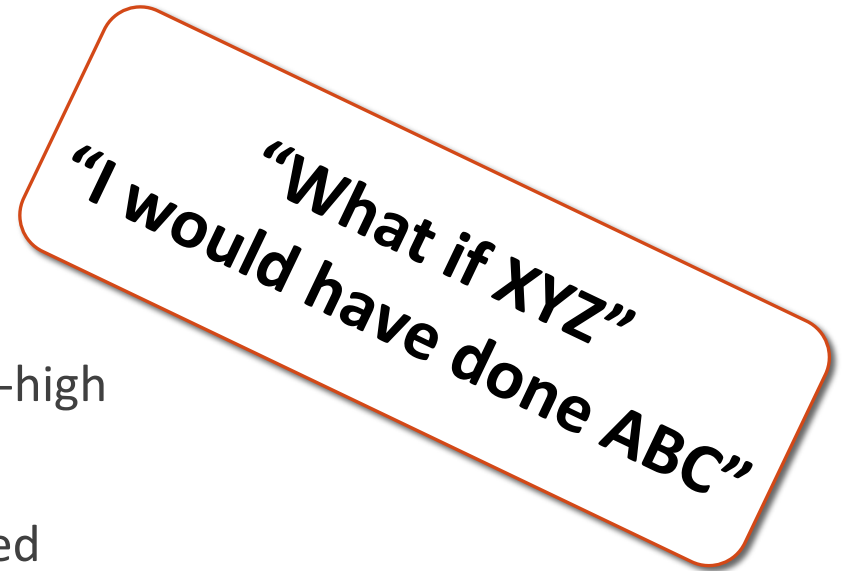


Have you ever wished for a time-machine?

- **Ctrl+Z** for:

- Contaminated cell cultures
- Words that hurt your loved ones
- Sunscreen you forgot at home before going to the beach
- One sell-it-all decision before the stock market went rocket-high
- A haircut choice that was too spontaneous
- An almost-finished thesis went unsaved when laptop crashed
- Other important life decisions...



Git(Hub)

PYTHON COURSE 2025-6

What is Git(Hub)?

Git tracks **versions of files**.

- Open source
- distributed version control system
- Creator: Linus Torvald for Linux development

GitHub is the platform that implement Git and host **those files**.

- Other platform: **GitLab** etc.



Btw, what or who is this?



[hands-on] Github account registration

github.com

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

[hands-on] create one GitHub repository

github.com

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

[hands-on] shield your access to Github

SSH

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

[hands-on] download your repo to BWJupyter

git clone

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

[hands-on] modify something

```
touch whatever.txt
```

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

[hands-on] view your changes

git status

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

[hands-on] STAGE your changes

git add FILENAME

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

Changed, staged.



Tracked, changed, unstaged



Untracked



[hands-on] view your changes 2

git diff

Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat

Changed, staged.



Tracked, changed, unstaged



Untracked



[hands-on] commit your changes

```
git commit -m "MSG"
```

Cheat sheet:

https://yopad.eu/p/Quick_Git_Chat

Changed, staged.



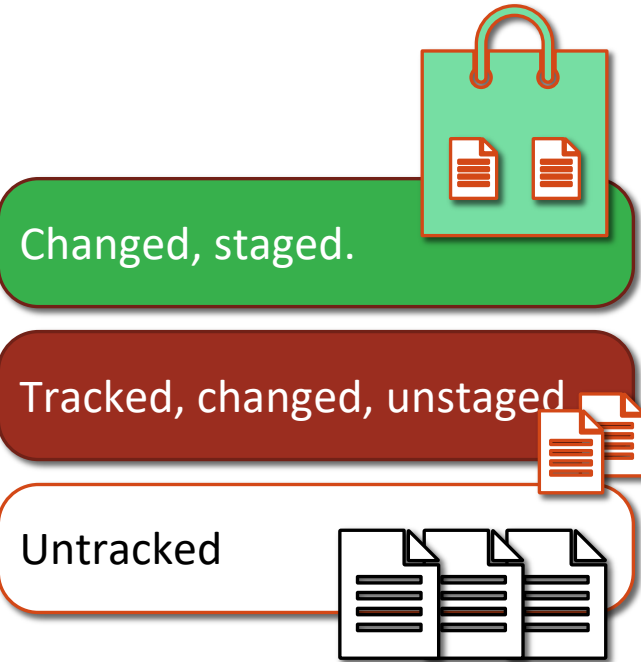
Tracked, changed, unstaged



Untracked



WHY?



[hands-on] rewind/regret

git log

git show HASH

git show HASH:PATH_TO_FILE

Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat

Changed, staged.



Tracked, changed, unstaged



Untracked



[hands-on] rewind/regret

```
git restore —source HASH:PATH_TO_FILE
```

Changed, staged.



Tracked, changed, unstaged



Untracked



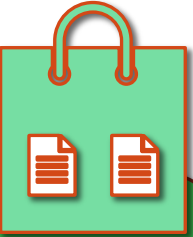
Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat

Are you happy with the changes now?



git push

Cheat sheet:
https://yopad.eu/p/Quick_Git_Chart



Changed, staged.

Tracked, changed, unstaged



Untracked



[hands-on] Practice

- Download one Github repo
- Navigate to one folder
- Answer one question
- Commit your answer

Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat

Changed, staged.



Tracked, changed, unstaged



Untracked





Branch

Git: when it's no longer one-man-ship

```
git branch -C branch_FIRSTNAME
```

Changed, staged.



Tracked, changed, unstaged



Untracked



Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat

[hands-on] make your branch visible to teammates

```
git push -u origin YOUR_BRANCH_NAME
```

Changed, staged.

Tracked, changed, unstaged

Untracked

Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat



[hands-on] make your branch visible to teammates

```
git push -u origin YOUR_BRANCH_NAME
```

Changed, staged.

Tracked, changed, unstaged

Untracked

Cheat sheet:
https://yopad.eu/p/Quick_Git_Chat



INSTALLATION & GUIs

With platform specific installers for Git, GitHub also provides the ease of staying up-to-date with the latest releases of the command line tool while providing a graphical user interface for day-to-day interaction, review, and repository synchronization.

GitHub for Windows
https://windows.github.com

GitHub for Mac
https://mac.github.com

For Linux and Solaris platforms, the latest release is available on the official Git web site.

Git for All Platforms
http://git-scm.com

SETUP

Configuring user information used across all local repositories

git config --global user.name "[firstname lastname]"
set a name that is identifiable for credit when review version history
git config --global user.email "[valid-email]"
set an email address that will be associated with each history marker
git config --global color.ui auto
set automatic command line coloring for Git for easy reviewing

SETUP & INIT

Configuring user information, initializing and cloning repositories

git init
initialize an existing directory as a Git repository
git clone [url]
retrieve an entire repository from a hosted location via URL

STAGE & SNAPSHOT

Working with snapshots and the Git staging area

git status
show modified files in working directory, staged for your next commit
git add [file]
add a file as it looks now to your next commit (stage)
git reset [file]
unstage a file while retaining the changes in working directory
git diff
diff of what is changed but not staged
git diff --staged
diff of what is staged but not yet committed
git commit -m "[descriptive message]"
commit your staged content as a new commit snapshot

BRANCH & MERGE

Isolating work in branches, changing context, and integrating changes

git branch
list your branches. a * will appear next to the currently active branch
git branch [branch-name]
create a new branch at the current commit
git checkout
switch to another branch and check it out into your working directory
git merge [branch]
merge the specified branch's history into the current one
git log
show all commits in the current branch's history

INSPECT & COMPARE

Examining logs, diffs and object information

git log
show the commit history for the currently active branch
git log branchB..branchA
show the commits on branchA that are not on branchB
git log --follow [file]
show the commits that changed file, even across renames
git diff branchB...branchA
show the diff of what is in branchA that is not in branchB
git show [SHA]
show any object in Git in human-readable format

TRACKING PATH CHANGES

Versioning file removes and path changes

git rm [file]
delete the file from project and stage the removal for commit
git mv [existing-path] [new-path]
change an existing file path and stage the move
git log --stat -M
show all commit logs with indication of any paths that moved

IGNORING PATTERNS

Preventing unintentional staging or committing of files

logs/ *.notes pattern*/
Save a file with desired patterns as .gitignore with either direct string matches or wildcard globs.
git config --global core.excludesfile [file]
system wide ignore pattern for all local repositories

SHARE & UPDATE

Retrieving updates from another repository and updating local repos

git remote add [alias] [url]
add a git URL as an alias
git fetch [alias]
fetch down all the branches from that Git remote
git merge [alias]/[branch]
merge a remote branch into your current branch to bring it up to date
git push [alias] [branch]
Transmit local branch commits to the remote repository branch
git pull
fetch and merge any commits from the tracking remote branch

REWRITE HISTORY

Rewriting branches, updating commits and clearing history

git rebase [branch]
apply any commits of current branch ahead of specified one
git reset --hard [commit]
clear staging area, rewrite working tree from specified commit

TEMPORARY COMMITS

Temporarily store modified, tracked files in order to change branches

git stash
Save modified and staged changes
git stash list
list stack-order of stashed file changes
git stash pop
write working from top of stash stack
git stash drop
discard the changes from top of stash stack

GitHub tutorials

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Learn how to start building, shipping, and maintaining software with GitHub. Explore our products, sign up for an account, and connect with the world's largest development community.

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[About GitHub and Git](#)

You can use GitHub and Git to collaborate on work.

[GitHub's plans](#)

An overview of GitHub's pricing plans.

[Getting started with your GitHub account](#)

With a personal account on GitHub, you can import or create repositories, collaborate with others, and connect with the GitHub community.

[Getting started with GitHub Team](#)

With GitHub Team groups of people can collaborate across many projects at the same time in an organization account.

Popular

[Creating an account on GitHub](#)

Create a personal account to get started with GitHub.

[Hello World](#)

Follow this Hello World exercise to learn GitHub's pull request workflow.

[Set up Git](#)

At the heart of GitHub is an open-source version control system (VCS) called Git. Git is responsible for everything GitHub-related that happens locally on your computer.

[About versions of GitHub Docs](#)

You can read documentation that reflects the GitHub product you're currently using.